

GreenMind AI – Intelligent Multi-Environmental Monitoring Recommendation System

GreenMind AI is an AI-powered decision support system developed for the Debrecen Green Sentinel environmental monitoring challenge. Instead of only visualizing existing environmental measurements, it recommends where Debrecen should expand its environmental monitoring network by identifying locations with limited monitoring coverage and high environmental importance.

The system combines the official 30-day Green Sentinel dataset with DKV public transport data. During preprocessing, air, noise and groundwater measurements are standardized into a common format, invalid values are converted to missing values, duplicate records are removed, timestamps are validated and measurements are rounded for consistent analysis. DKV passenger statistics are also processed to estimate transport activity across the city.

To estimate conditions at locations without sensors, GreenMind AI applies Inverse Distance Weighting (IDW) interpolation using nearby monitoring stations. The model estimates air quality (PM_{2.5}, PM₁₀, NO₂ and O₃), wind speed, environmental variability, noise levels and groundwater conditions for every candidate location.

Each candidate receives three independent suitability scores: **Air Suitability**, **Noise Suitability** and **Groundwater Suitability**. Air suitability considers monitoring coverage, pollution, historical variability and wind conditions. Noise suitability combines monitoring coverage, estimated noise and nearby DKV activity, while groundwater suitability evaluates monitoring coverage and groundwater conditions. These are combined into a single Priority Score that ranks the best locations for installing a new Green Sentinel station. Every recommendation also includes a confidence score, helping users understand how reliable the prediction is.

A key feature is the interactive simulation module. Users can virtually add a recommended station, after which the FastAPI backend recalculates coverage, environmental estimates, suitability scores and recommendations in real time. This allows planners to compare different expansion strategies before installing physical infrastructure.

The application follows a backend-driven architecture with a React frontend, FastAPI backend and spatial recommendation engine. It also includes an environmental monitoring dashboard, data-quality assessment and a methodology page explaining the preprocessing, interpolation and scoring process. By combining air quality, noise, groundwater and transport information, GreenMind AI helps municipalities reduce monitoring blind spots, prioritize sensor investments and support evidence-based urban planning.